SY B.Sc. Data Science

Unit II – Introduction to Data Mining

Dr. Savita Mohurle

The data is acknowledging those conveys certain information in the form of character, numbers, and symbols and have certain meaning. The data is quantitative and qualitative in nature on which certain operations can be performed using computer.

The data is one which can be transferred verbally, non-verbally and in the form of electronic signals.

Knowledge is a form of awareness or familiarity. It is often understood as awareness of facts or as practical skills, and may also mean familiarity with objects or situations.

Information is an abstract concept that refers to that which has the power to inform.

At the most fundamental level, information pertains to the interpretation (perhaps formally) of that which may be sensed, or their abstractions.

A pattern is a series of data that repeats in a recognizable way.

It can be identified in the history of the asset being evaluated or other assets with similar characteristics.

Patterns often include the study of sale volume, as well as price.

A relationship refers to the correspondence between two variables. When we talk about types of relationships, we can mean that in at least two ways: the nature of the relationship or the pattern of it.

What is Data Mining?

- Data mining is one of the most useful techniques that help entrepreneurs, researchers, and individuals to extract valuable information from huge sets of data.
- Data mining is the process of extracting useful information from an accumulation of data, often from a data warehouse or collection of linked data sets.
- Data Mining is the process of investigating hidden patterns of information to various perspectives for categorization into useful data, which is collected and assembled in particular areas such as data warehouses, efficient analysis, data mining algorithm, helping decision making and other data requirement to eventually cost-cutting and generating revenue.

What is Data Mining?

- Data mining is the act of automatically searching for large stores of information to find trends and patterns that go beyond simple analysis procedures.
- Data Mining is a process used by organizations to extract specific data from huge databases to solve business problems. It primarily turns raw data into useful information.
- Data mining is also called ***Knowledge Discovery in Database (KDD)***.

- Why use data mining? The primary benefit of data mining is its power to identify patterns and relationships in large volumes of data from multiple sources.
- Why should we need data mining? It helps data scientists easily analyze enormous amounts of data quickly. Data scientists can use the information to detect fraud, build risk models, and improve product safety. It helps data scientists quickly initiate automated predictions of behaviors and trends and discover hidden patterns.

Uses of Data Mining

- Data mining is the process of gathering information and analyzing it for actionable patterns, which can then be used to develop marketing strategies, new products that fit customers' wants and needs, and cost-saving strategies.

- Basket Analysis
- Sales Forecasting
- Database Marketing
- Inventory Planning
- Customer Loyalty

Data Mining Techniques

- Association rules
- Clustering
- Classification
- Decision tree
- Neural network
- Predictive analysis
- KNN



The world is data rich but information poor.

The widening gap between data and information calls for the systematic development of data mining tools that can turn data tombs into "golden nuggets" of knowledge.

In addition, many other terms have a similar meaning to data mining—for example, knowledge mining from data, knowledge extraction, data/pattern analysis, data archaeology, and data dredging.

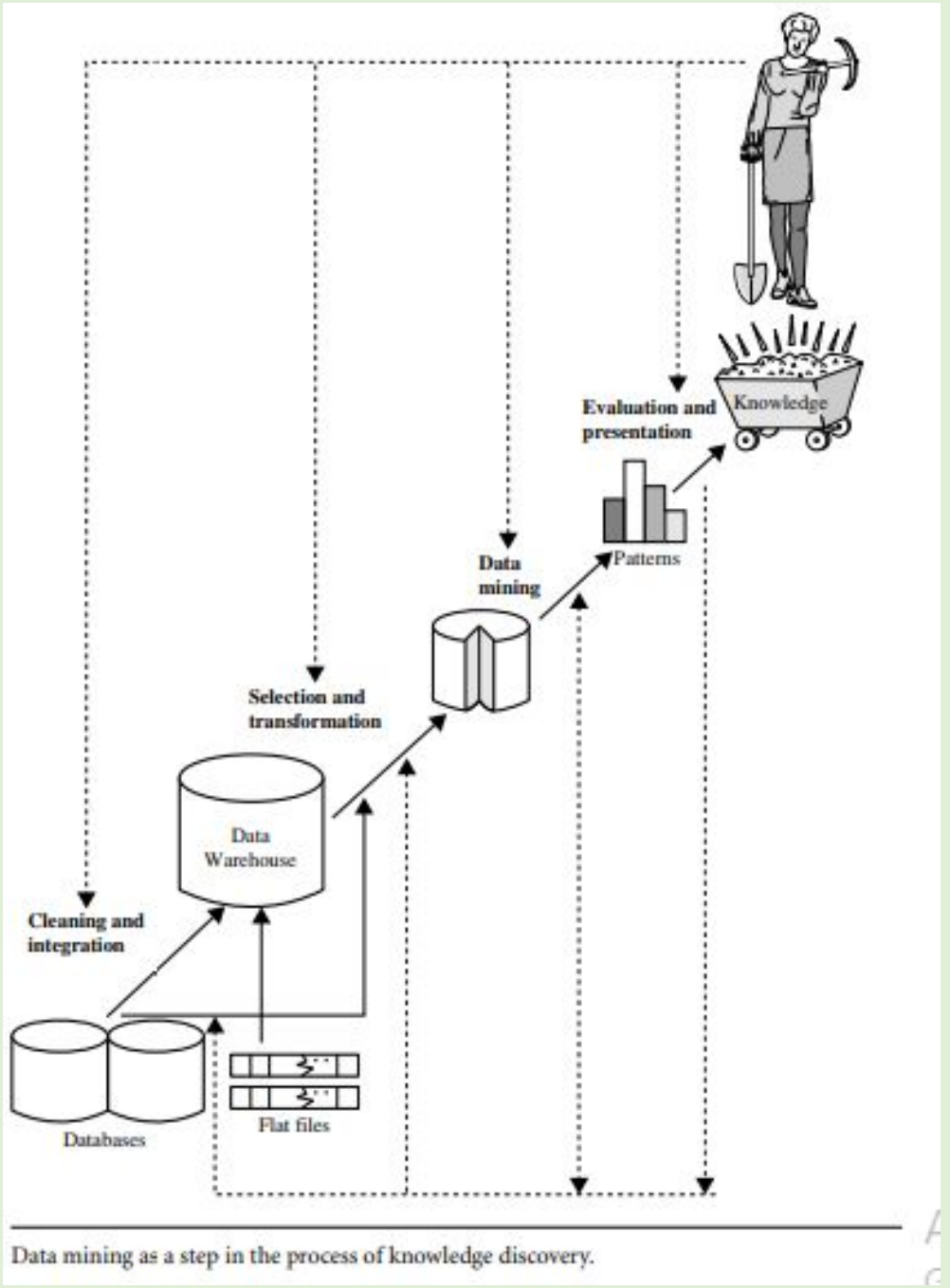


Data mining—searching for knowledge (interesting patterns) in data.

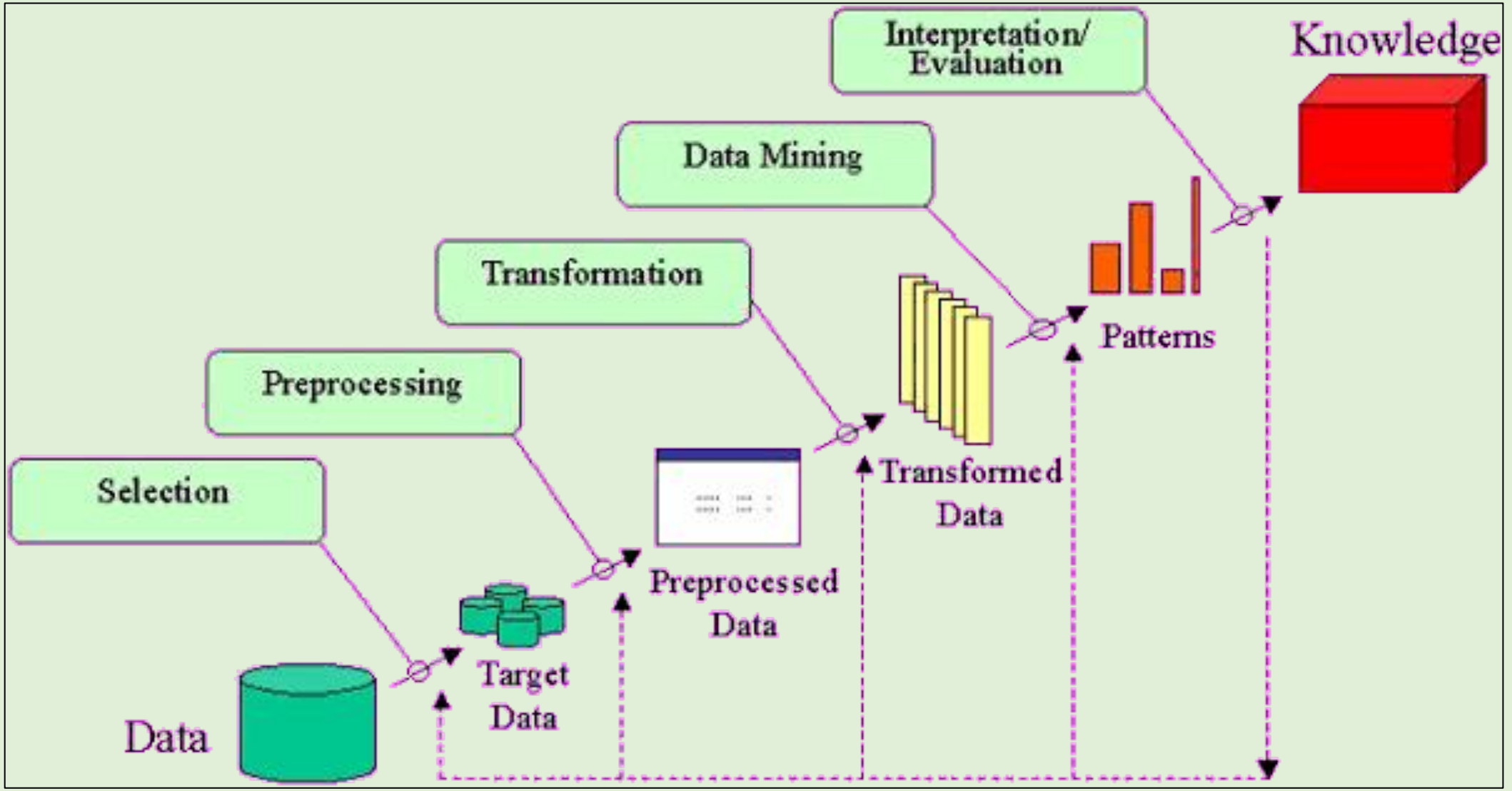
Knowledge Discovery Process

The knowledge discovery process is an iterative sequence of the following steps:

- **Data cleaning** (to remove noise and inconsistent data)
- **Data integration** (where multiple data sources may be combined)
- **Data selection** (where data relevant to the analysis task are retrieved from the database)
- **Data transformation** (where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations)
- **Data mining** (an essential process where intelligent methods are applied to extract data patterns) 6. Pattern evaluation (to identify the truly interesting patterns representing knowledge based on interestingness measures)
- **Knowledge presentation** (where visualization and knowledge representation techniques are used to present mined knowledge to users)



Data mining as a step in the process of knowledge discovery.


Interpretation/
Evaluation
Knowledge
Data Mining
Transformation
Preprocessing
Selection
Patterns
Transformed
Data
Preprocessed
Data
Target
Data
Data

What Kinds of Data Can Be Mined?

- As a general technology, data mining can be applied to any kind of data as long as the data are meaningful for a target application.
- The most basic forms of data for mining applications are:

 Database data (DBMS and Relational DBMS) - When **mining relational databases**, we can go further by *searching for trends* or *data patterns*. For example, data mining systems can analyze customer data to predict the credit risk of new customers based on their income, age, and previous credit information.

 Data warehouse data

 Transactional data - Transaction data or transaction information is a category of data describing transactions. Transaction data/information gather variables generally referring to reference data or master data – e.g. dates, times, time zones, currencies.

 Data mining can also be applied to other forms of data (e.g., data streams, ordered/sequence data, graph or networked data, spatial data, text data, multimedia data, and the WWW).

What Kinds of Patterns Can Be Mined?

- There are a number of data mining functionalities included:

- Characterization and discrimination
- Mining of frequent patterns, associations, and correlations
- Classification and regression
- Clustering analysis
- Outlier analysis

In general, such tasks can be classified into two categories:

- Descriptive mining - tasks characterize properties of the data in a target data set.
- Predictive mining tasks perform induction on the current data in order to make predictions.

"Interesting patterns represent knowledge."

Methods to Detect Outliers

1. Statistical Methods

A. Z-Score Method

Formula:

$$Z = \frac{X - \mu}{\sigma}$$

If |Z| > 3, it's usually an outlier.

Methods to Detect Outliers

2. IQR (Interquartile Range) Method

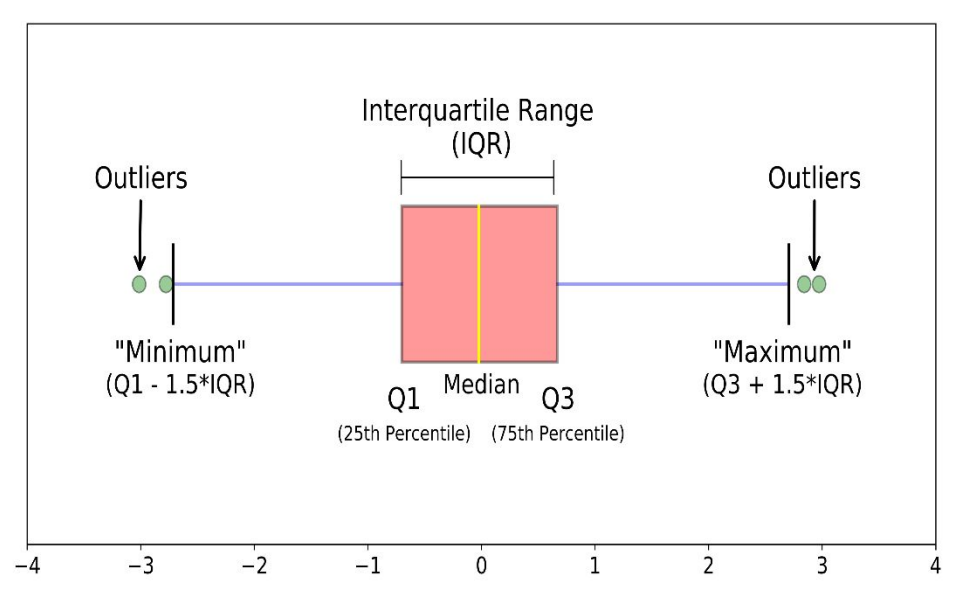
Find Q1 (25%) and Q3 (75%)

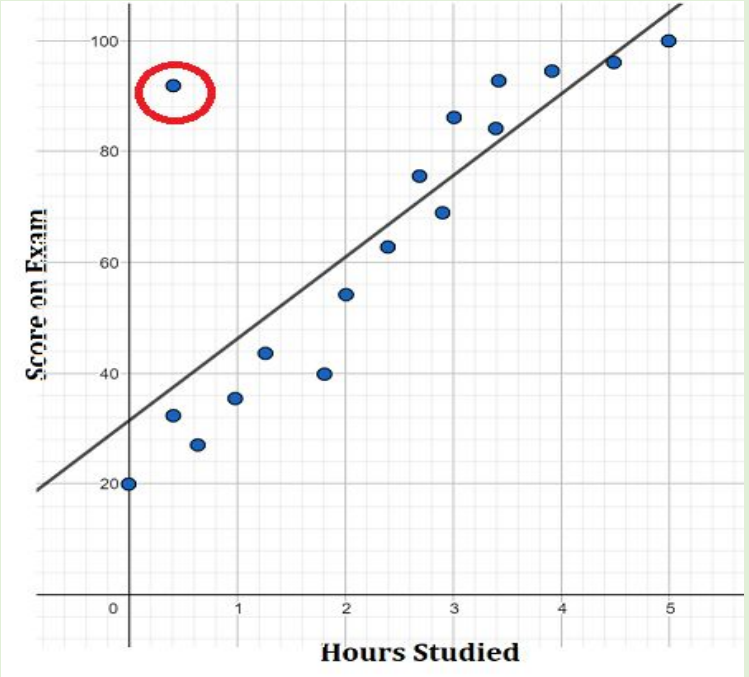
IQR = Q3 − Q1

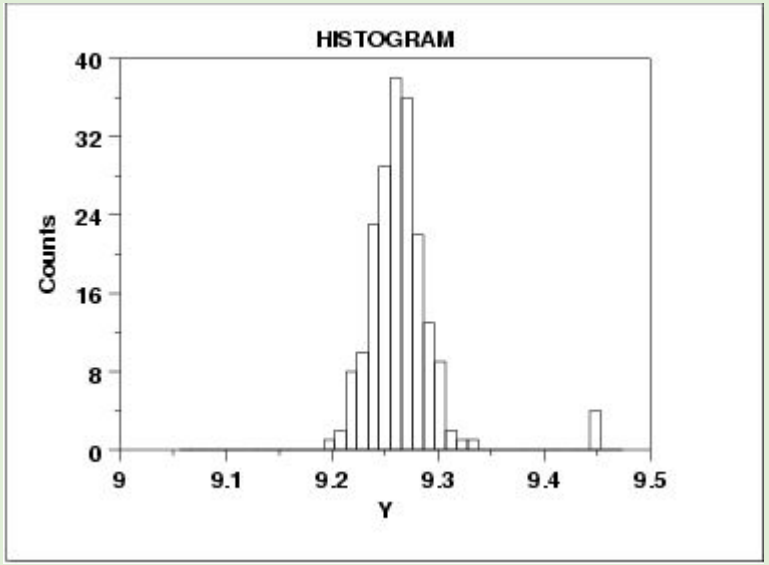
Outlier limits:

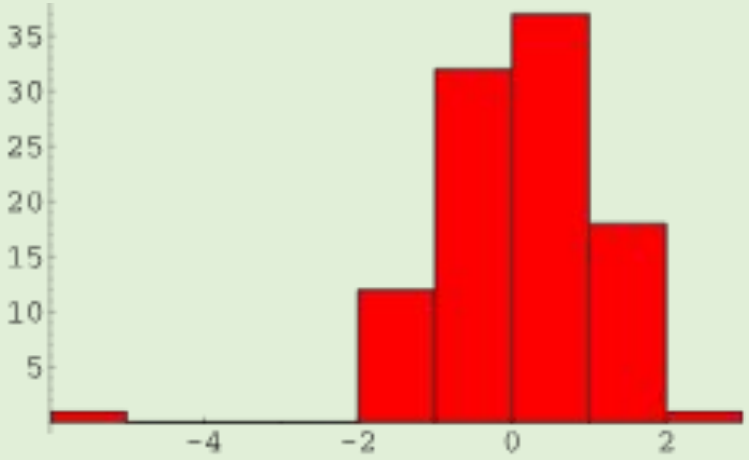
Lower Bound = Q1 − 1.5 × IQR
Upper Bound = Q3 + 1.5 × IQR

Methods to Detect Outliers







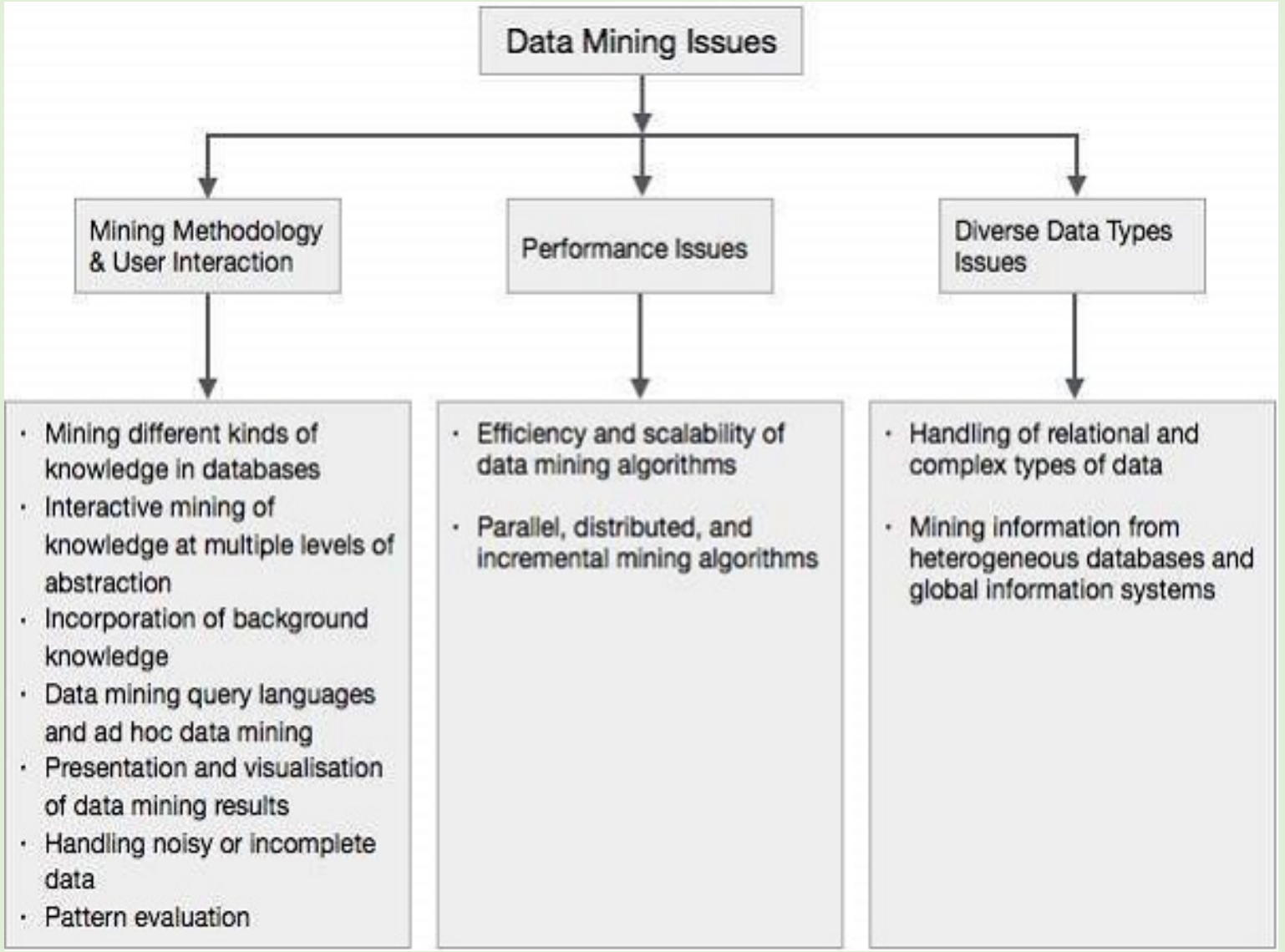


Methods to Detect Outliers

3. Machine Learning Method

Method	**Use Case**
Isolation Forest	Large datasets
DBSCAN	Density-based detection
One-Class SVM	Novelty detection
LOF (Local Outlier Factor)	Detects local density differences

Data mining Issues



Data mining Applications

- Market Analysis
- Fraud Detection
- Customer Retention
- Production Control
- Science Exploration
- Financial Data Analysis
- Retail Industry
- Telecommunication Industry
- Biological Data Analysis
- Other Scientific Applications
- Intrusion Detection

Data Objects and Attribute Types

Data sets are made up of data objects.

- A **data object** represents an entity—in a sales database, the objects may be customers, store items, and sales; in a medical database, the objects may be patients; in a university database, the objects may be students, professors, and courses.
- Data objects are typically described by attributes.
- Data objects can also be referred to as *samples, examples, instances, data points*, or *objects*.
- If the data objects are stored in a database, they are *data tuples*. That is, the rows of a database correspond to the data objects, and the columns correspond to the attributes.

Data Objects and Attribute Types

- An attribute is a data field, representing a characteristic or feature of a data object. The terms attribute, dimension, feature, and variable are often used interchangeably.
- Attributes describing a customer object can include, for example, customer ID, name, and address
- A set of attributes used to describe a given object is called an attribute vector (or feature vector).
- The distribution of data involving one attribute (or variable) is called univariate.
- A bivariate distribution involves two attributes, and so on.
- The type of an attribute is determined by the set of possible values—nominal, binary, ordinal, or numeric

Nominal Attributes

- Nominal means “relating to names.”
- The values of a nominal attribute are symbols or names of things.
- Each value represents some kind of category, code, or state, and so nominal attributes are also referred to as categorical.
- The values do not have any meaningful order.
- In computer science, the values are also known as enumerations

Suppose that hair color and marital status are two attributes describing person objects. In our application, possible values for hair color are black, brown, blond, red, auburn, gray, and white. The attribute marital status can take on the values single, married, divorced, and widowed. Both hair color and marital status are nominal attributes. Another example of a nominal attribute is occupation, with the values teacher, dentist, programmer, farmer, and so on

Binary Attributes

- A binary attribute is a nominal attribute with only two categories or states: 0 or 1, where 0 typically means that the attribute is absent, and 1 means that it is present.
- Binary attributes are referred to as Boolean if the two states correspond to true and false.

Given the attribute smoker describing a patient object, 1 indicates that the patient smokes, while 0 indicates that the patient does not. Similarly, suppose the patient undergoes a medical test that has two possible outcomes. The attribute medical test is binary, where a value of 1 means the result of the test for the patient is positive, while 0 means the result is negative

A binary attribute is symmetric if both of its states are equally valuable and carry the same weight; that is, there is no preference on which outcome should be coded as 0 or 1. One such example could be the attribute gender having the states male and female.

A binary attribute is asymmetric if the outcomes of the states are not equally important, such as the positive and negative outcomes of a medical test for HIV. By convention, we code the most important outcome, which is usually the rarest one, by 1 (e.g., HIV positive) and the other by 0 (e.g., HIV negative).

Ordinal attribute

- An ordinal attribute is an attribute with possible values that have a meaningful order or ranking among them, but the magnitude between successive values is not known.

Grade (e.g., A+, A, A−, B+, and so on) and professional rank. Professional ranks can be enumerated in a sequential order: for example, assistant, associate, and full for professors, and private, private first class, specialist, corporal, and sergeant for army ranks.

Customer satisfaction had the following ordinal categories: 0: very dissatisfied, 1: somewhat dissatisfied, 2: neutral, 3: satisfied, and 4: very satisfied.

If integers are used, they represent computer codes for the categories, as opposed to measurable quantities (e.g., 0 for small drink size, 1 for medium, and 2 for large).

Numeric Attributes

- A numeric attribute is quantitative; that is, it is a measurable quantity, represented in integer or real values.
- Numeric attributes can be interval-scaled or ratio-scaled.

 Interval-Scaled Attributes

 Ratio-Scaled Attributes

Numeric Attributes

- Interval-scaled attributes are measured on a scale of equal-size units. The values of interval-scaled attributes have order and can be positive, 0, or negative.
- Thus, in addition to providing a ranking of values, such attributes allow us to compare and quantify the difference between values.

A temperature attribute is interval-scaled. Because interval-scaled attributes are numeric, we can compute their mean value, in addition to the median and mode measures of central tendency

- A ratio-scaled attribute is a numeric attribute with an inherent zero-point.
- That is, if a measurement is ratio-scaled, we can speak of a value as being a multiple (or ratio) of another value.
- In addition, the values are ordered, and we can also compute the difference between values, as well as the mean, median, and mode

Examples include attributes to measure weight, height, latitude and longitude coordinates (e.g., when clustering houses), and monetary quantities (e.g., you are 100 times richer with $100 than with $1).

Nominal	Symbols or names (or numbers) of things	Occupation (teacher, dentist, programmer, farmer, and so on)	**Qualitative**
Binary	A nominal attribute with only two categories or states	Symmetric - if both of its states are equally valuable and carry the same weight; Example: 0 or 1, true or false, male or female, yes or no	
		Asymmetric if the outcomes of the states are not equally important, such as the positive and negative outcomes of a medical test for HIV.	
Ordinal	An attribute with possible values that have a meaningful order or ranking among them, but the magnitude between successive values is not known.	Example: Small, medium, large	
Numeric	It is a measurable quantity, represented in integer or real values.	Interval scaled - Are measured on a scale of equal-size units Example: A temperature attribute is interval-scaled.	Quantitative
		Ratio scaled - Is a numeric attribute with an inherent zero-point. Example: height, money, age, weight, speed, and time periods.	

Statistical Description of Data

- Science of collecting, organizing, presenting, interpreting and analyzing.
- Data scientist is a person who is better at statistics than any programmer and better at programming than any statistician.

Descriptive statistics summarizes or describes the characteristics of a data set.

- Descriptive statistics consists of three basic categories of measures: measures of central tendency, measures of variability (or spread), and frequency distribution.

❑ Measures of central tendency describe the center of the data set (mean, median, mode).

❑ Measures of variability describe the dispersion of the data set (variance, standard deviation).

❑ Measures of frequency distribution describe the occurrence of data within the data set (count, frequency table, histogram, bar chart).

Statistical Description of Data

Example: Mean, Median, and Mode
Consider the dataset:
Data: 2, 4, 6, 8, 10, 10, 12

1. Mean (Average)

Formula:

$$\text{Mean} = \frac{\textit{Sum of all values}}{\textit{Number of values}}$$

$$2 + 4 + 6 + 8 + 10 + 10 + 12 = 52$$

Mean=752=7.43

Statistical Description of Data

Step 1: Arrange data (already sorted)

2, 4, 6, 8, 10, 10, 12

Step 2: Find position

$$\text{Median } position = \frac{n+1}{2} = \frac{7+1}{2} = 4$$

The 4th value = 8

Statistical Description of Data

Value	Frequency
2	1
4	1
6	1
8	1
10	2
12	1

Descriptive Analysis

Elements	Std Dev.	Mean	Median	Mode	Q1 (25%)	Q2 (50%)	Q3 (75%)	Minimum	Maximum
Moisture	7.936389	19.12252	19.0652	23.1	14.18	19.065	22.56	0	56.5
N	2.828273	1.153824	0.78	0.78	0.5	0.78	1.12	0	68
P205	1.147476	1.625204	1.43	0.95	0.8975	1.43	1.9	0.099	9.25
K20	2.440252	2.624122	1.59	0.98	0.97	1.59	3.88	0	19.8
Cd	2.120466	4.282424	4.12	5.1	2.99	4.12	5.34	0	16.2
C	8.628163	20.37754	17.4	16.34	15.7	17.4	22.1	0	67.89
C:N	8.658163	20.32102	19.66	20.1	16.873	19.66	20.873	2.3	56.2
Zn	144.6234	993.5245	1000	999	965	1000	1020.5	189.6	1500
Pb	14.16045	99.29923	99.89	100.1	98.665	99.89	101	0	267
Cu	64.09558	278.225	299.1	0	281.26	299.1	302	0	401.1
pH	1.718799	7.217256	7.32	7.23	6.5	7.32	8.1	0.5	24
Density	2.226921	1.76008	0.7675	0.45	0.42	0.7675	2.11	0.001	15.6
Conductivity	2.7050982	5.5016157	5.45	6.45	3.435	5.45	7.4574	0	19.2
As	2.026374	7.672502	7.45	6.55	7.45	8.5	0.5	0	24
Ni	16.34339	22.74485	18.685	4.5	9.2415	18.685	33.223	1.2	91.34
Hg	0.37413	0.184395	0.023	0.01	0.01	0.023	0.128	0	3.2
Cr	15.62833	20.15139	14.7	4.5	7.89	14.7	31.32	0	66.4

Statistical Description of Data

- *Inferential statistics helps study a sample of data and make conclusions about its population.*
- *A sample is a smaller data set drawn from a larger data set called the population.*
- *If the sample does not represent the population, one cannot make accurate estimations related to the latter.*
- *The purpose of studying inferential statistics is to infer the behavior of a population.*
- The accuracy of inferential statistics depends largely on the accuracy of sample data and how it represents the larger population.
- This can be effectively done by obtaining a random sample. Results that are based on non-random samples are usually discarded.

	Descriptive Statistics	Inferential Statistics
Purpose	Describe and summarize data	Make inferences and draw conclusions about a population based on sample data
Data Analysis	Analyzes and interprets the characteristics of a dataset	Uses sample data to make generalizations or predictions about a larger population
Population Vs Sample	Focuses on the entire population or dataset	Focuses on a subset of the population (sample) to draw conclusions about the entire population
Measurements	Provides measures of central tendency and dispersion	Estimates parameters, tests hypotheses, and determines the level of confidence or significance in the results
Examples	Mean, median, mode, standard deviation, range, frequency tables	Hypothesis testing, confidence intervals, regression analysis, ANOVA (analysis of variance), chi-square tests, t-tests, etc.
Goal	Summarize, organize, and present data	Generalize findings to a larger population, make predictions, test hypotheses, evaluate relationships, and support decision-making
Population Parameters	Not typically estimated	Estimated using sample statistics (e.g., sample mean as an estimate of population mean)
Sample Representativeness	Not required	Crucial; the sample should be representative of the population to ensure accurate inferences

- The mode is another measure of central tendency.
- The mode for a set of data is the value that occurs most frequently in the set.
- Therefore, it can be determined for qualitative and quantitative attributes.
- It is possible for the greatest frequency to correspond to several different values, which results in more than one mode.
- Data sets with one, two, or three modes are respectively called unimodal, bimodal, and trimodal.
- In general, a data set with two or more modes is multimodal.
- At the other extreme, if each data value occurs only once, then there is no mode.

- The range in statistics for a given data set is the difference between the highest and lowest values.
- For example, if the given data set is {2,5,8,10,3}, then the range will be 10 – 2 = 8.

- In statistics, **Quartiles** are the set of values which has three points dividing the data set into four identical parts. One of the best applications of quartiles is defined in box whisker plot.

 First quartile: 25% from smallest to largest of numbers

 Second quartile: between 25.1% and 50% (till median)

 Third quartile: 51% to 75% (above the median)

 Fourth quartile: 25% of largest numbers

Quartile deviation is defined as half of the distance between the third and the first quartile. It is also called **Semi Interquartile range**.

Quartile deviation = $(Q_3-Q_1)/2$

- The **interquartile range (IQR)** is the difference between the upper and lower quartile of a given data set and is also called a **midspread**. It is a measure of statistical distribution, which is equal to the difference between the upper and lower quartiles.

$IQR = Q_3 – Q_1$

Question 1: Find the quartiles of the following data: 4, 6, 7, 8, 10, 23, 34.

Solution: Here the numbers are arranged in the ascending order and number of items, n = 7

- Lower quartile, Q_1 = [(n+1)/4] th item
- Q_1= 7+1/4 = 2nd item = 6
- Median, Q_2 = [(n+1)/2]th item
- Q_2= 7+1/2 item = 4th item = 8
- Upper Quartile, Q_3 = [3(n+1)/4]th item
- Q_3 = 3(7+1)/4 item = 6th item = 23

What is data visualization?

- Data visualization is the representation of data through use of common graphics, such as charts, plots, infographics, and even animations. These visual displays of information communicate complex data relationships and data-driven insights in a way that is easy to understand.

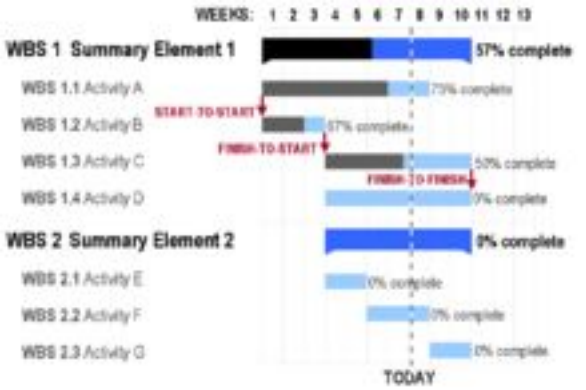
To help :

- Visually explore and discover
- Quickly understand
- Interpret and gain important insights into otherwise difficult-to-identify structures
- Relationships
- Correlations
- Local and global patterns
- Trends
- Variations
- Constancy, clusters, outliers and unusual groupings within data (exploratory visualization)

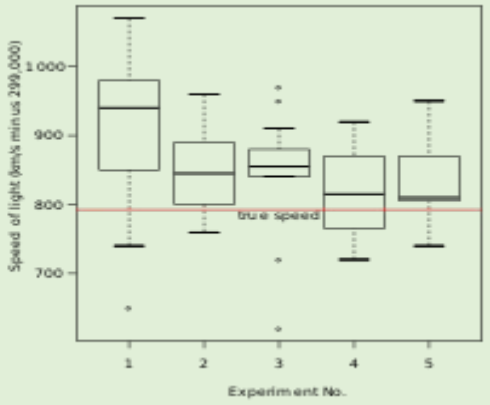
What is data visualization?

The visual formats used in data visualization include:

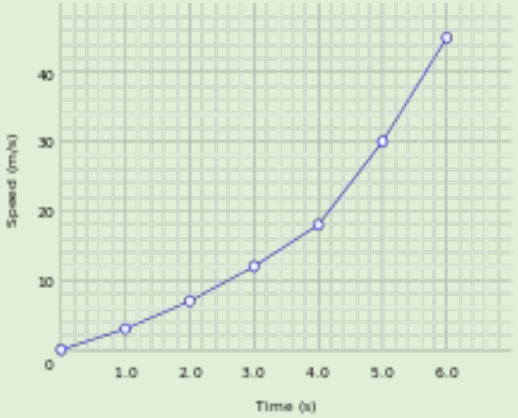
- Tables
- Charts and graphs (Example: pie chart, bar chart, line chart, arear chart, cone chart, pyramid chart, donut chart, histograms, spectrograms, cohort charts, waterfall chart, funnel chart, bullet chart, etc.)
- Diagrams
- Plots (Example: scatter plots, distribution plots, box-and-whisker plots)
- Geospatial maps(Example: proportional symbol maps, choropleth maps, isopleth maps, and heat maps)
- Figures
- Correlation matrices
- Percentage gauges



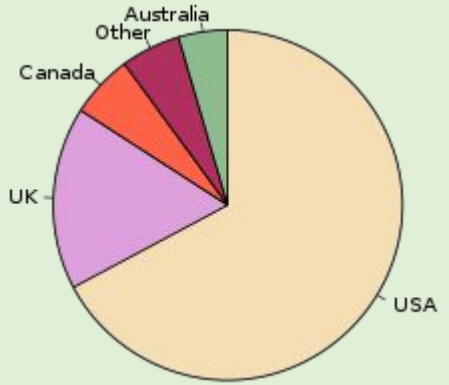
Gantt Chart



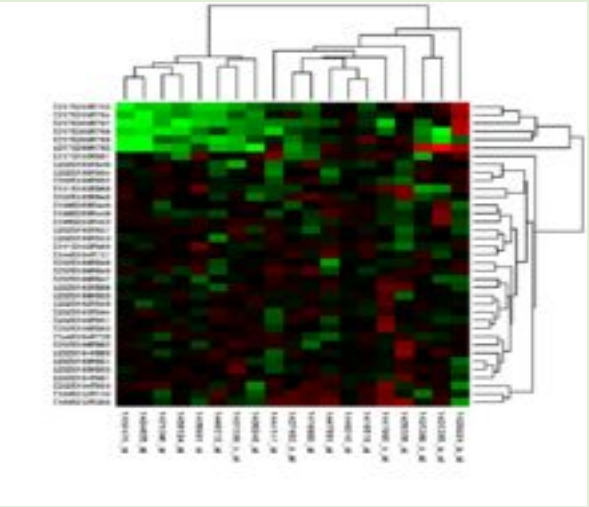
Box-and-Whisker Plot



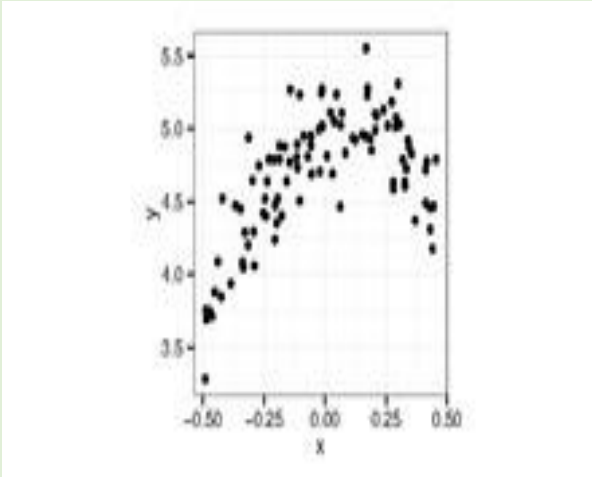
Line Graph



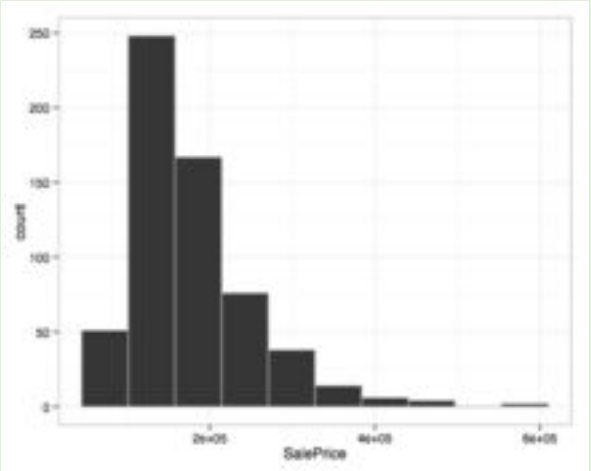
Pie Chart



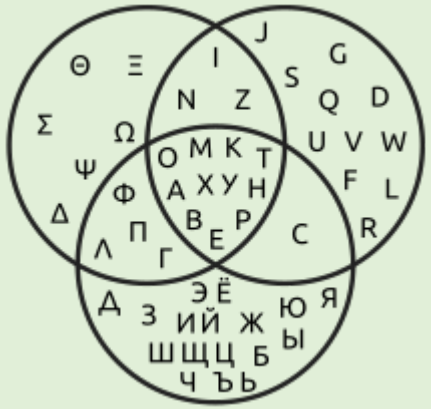
Heat Map



Scattered Plot



Bar Graph



Venn diagram

Data similarity and dissimilarity measures.

Similarity measure

- is a numerical measure of how alike two data objects are.
- higher when objects are more alike.
- often falls in the range [0,1]

Similarity might be used to identify

- duplicate data that may have differences due to typos.
- equivalent instances from different data sets. E.g. names and/or addresses that are the same but have misspellings.
- groups of data that are very close (clusters)

Dissimilarity measure

- is a numerical measure of how different two data objects are
- lower when objects are more alike
- minimum dissimilarity is often 0 while the upper limit varies depending on how much variation can be

Dissimilarity might be used to identify

- outliers
- interesting exceptions, e.g. credit card fraud
- boundaries to clusters

Proximity refers to either a similarity or dissimilarity

Data proximity is a data-driven approach to corpus semantics analysis in various data mining techniques, which discriminate the data based upon the idea that words occurring together or in similar contexts.

Proximity measures are mainly mathematical techniques that calculate the similarity/dissimilarity of data points in Data science.

Single attribute sim/dissim measures

Attribute Type	Dissimilarity	Similarity
Nominal	$d = \begin{cases} 0 & \text{if } x = y \\ 1 & \text{if } x \neq y \end{cases}$	$s = \begin{cases} 1 & \text{if } x = y \\ 0 & \text{if } x \neq y \end{cases}$
Ordinal	$d = \lvert x - y \rvert / (n-1)$ (values mapped to integers 0 to $n-1$, where n is the number of values)	$s = 1 - d$
Interval or Ratio	$d = \lvert x - y \rvert$	$s = -d$, $s = \frac{1}{1+d}$, $s = e^{-d}$, $s = 1 - \frac{d - min_d}{max_d - min_d}$

- **Nominal** is binary if two values are equal or not
- **Ordinal** is the difference between two values, normalized by the maximum distance
- **Quantitative** dissimilarity is just a distance between, similarity attempts to scale that distance to [0,1]

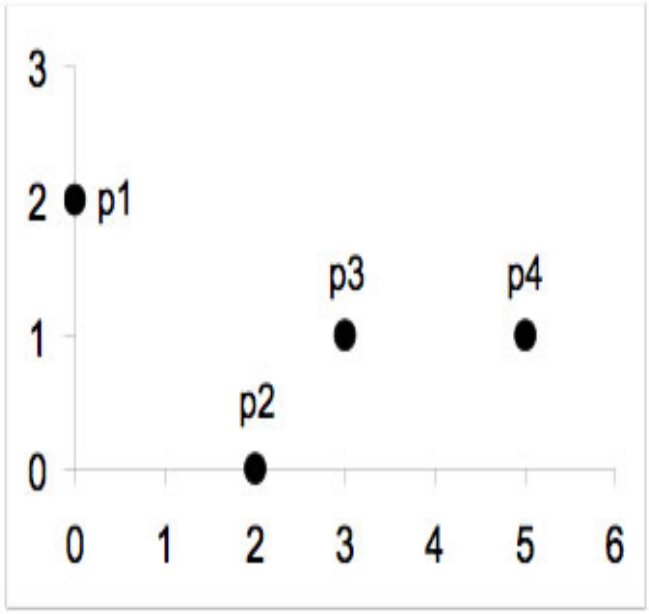
Distance between instances with multiple attributes.

Euclidean distance

$$d(\mathbf{x}, \mathbf{y}) = \sqrt{\sum_{k=1}^{n} (x_k - y_k)^2}$$

where *n* is the number of dimensions (attributes) and x_k and y_k are, respectively, the *k-th* attributes (components) or data objects **x** and **y**

Standardization/normalization may be necessary to ensure an attribute does not skew the distances due to different scales.



point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

Distance Matrix

Euclidean distance

Applying Pythagoras theorem for triangle ABC.

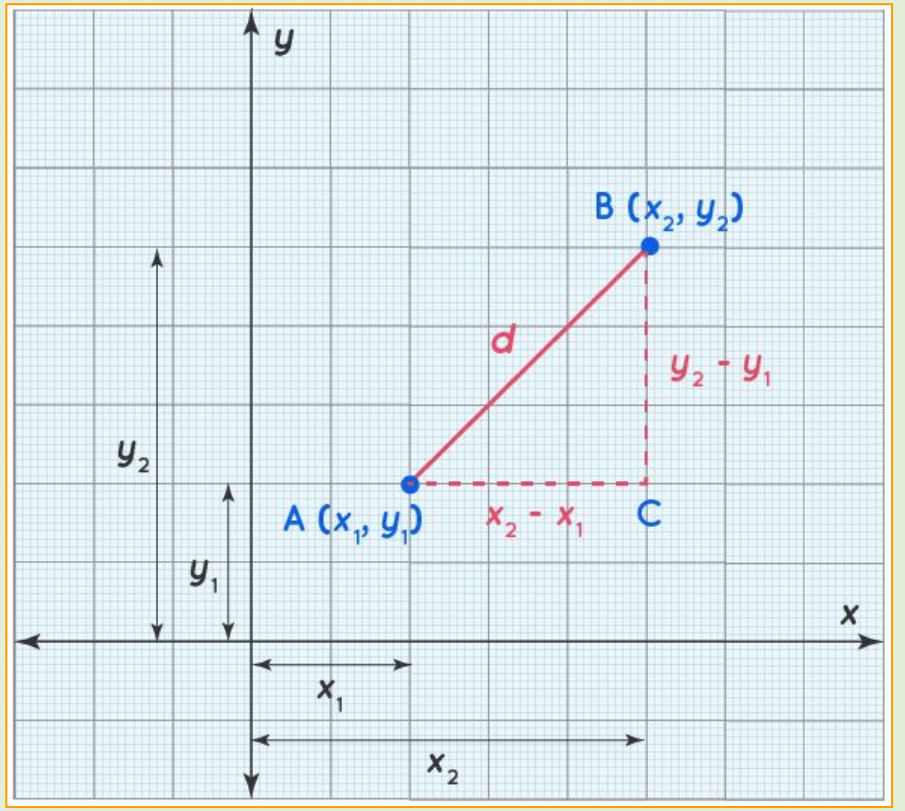
$\square\ AB^2 = AC^2 + BC^2$

$\square\ d^2 = (x22 - x11)^2 + (y22 - y11)^2$

Taking the square root on both sides,

$\square\ d = \sqrt{[\ (x22 - x11)^2 + (y22 - y11)^2]}$

Hence the Euclidean distance formula is derived.



Euclidean distance – Example 1

Find the Euclidean distance between two points (a, b) and (-a, -b).

Solution:

Let the point P be (a, b) and Q be (-a, -b)

i.e. $P(a, b) = (x_1, y_1)$ and $Q(-a, -b) = (x_2, y_2)$

We know that the Euclidean distance formula is,

Euclidean distance, $d = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

Now, substitute the values in the formula, we get

$d = \sqrt{[(-a - a)^2 + (-b - b)^2]}$

$d = \sqrt{[(-2a)^2 + (-2b)^2]}$

$d = \sqrt{[4a^2 + 4b^2]}$

$d = \sqrt{4(a^2+b^2)}$

$d = 2\sqrt{(a^2+b^2)}$.

Hence, the distance between two points (a, b) and (-a, -b) is $2\sqrt{(a^2+b^2)}$.

Euclidean distance – Example 2

Find the distance between two points P(0, 4) and Q(6, 2).

Solution:

Let P(0, 4) = (x_1, y_1) and Q(6, 2) = (x_2, y_2)

The distance between the point PQ is

$PQ = \sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$

$PQ = \sqrt{[(6 - 0)^2 + (2 - 4)^2]}$

$PQ = \sqrt{[(6)^2 + (-2)^2]}$

$PQ = \sqrt{(36+4)}$

$PQ = \sqrt{40}$

$PQ = 2\sqrt{10}$.

Therefore, the distance between two points P(0,4) and Q(6, 2) is $2\sqrt{10}$.

Distance between instances with multiple attributes

Minkowski Distance - is a generalization of Euclidean Distance

$$d(\mathbf{x}, \mathbf{y}) = \left(\sum_{k=1}^{n} |x_k - y_k|^r \right)^{1/r}$$

where *r* is a parameter, *n* is the number of dimensions (attributes) and x_k and y_k are, respectively, the *k-th* attributes (components) or data objects **x** and **y.**

Examples

r = 1. "City block", "Manhattan", "taxicab", L_1 norm distance.

- Another example of this is the Hamming distance, which is just the number of bits that are different between two binary vectors

r = 2. Euclidean distance (L_2 norm)

r = ∞. "supremum" (L_{max} norm, L_∞ norm) distance. This is the maximum difference between any component of the vectors

Do not confuse r with n, i.e., all these distance measures are defined for all numbers of dimensions.

point	x	y
p1	0	2
p2	2	0
p3	3	1
p4	5	1

L1	p1	p2	p3	p4
p1	0	4	4	6
p2	4	0	2	4
p3	4	2	0	2
p4	6	4	2	0

L2	p1	p2	p3	p4
p1	0	2.828	3.162	5.099
p2	2.828	0	1.414	3.162
p3	3.162	1.414	0	2
p4	5.099	3.162	2	0

L∞	p1	p2	p3	p4
p1	0	2	3	5
p2	2	0	1	3
p3	3	1	0	2
p4	5	3	2	0

Distance Matrix

Thank You